

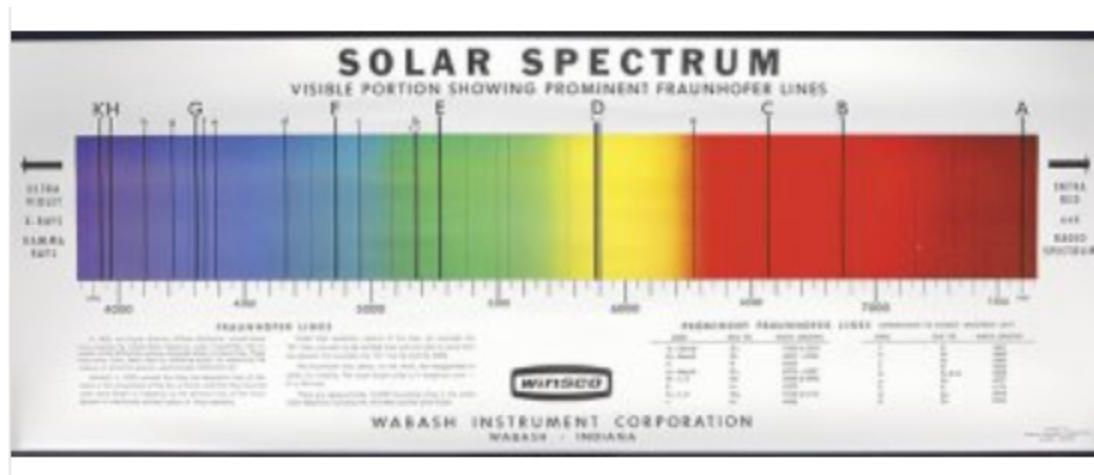
popquiz18 stellar parameters 2

⚠ This is a preview of the published version of the quiz

Started: Mar 27 at 4:37pm

Quiz Instructions

absorption lines are like QR codes.



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• •
• •

Question 1 10 pts

Who made the connection between temperature (of hot gas/stars) and absorption lines

(Bengali physicist)

- ☐ Niels Bohr
- ☐ Wilhelm Wien
- ☐ Max Planck
- ☐ Meghnad Saha

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Question 2 10 pts

Who came up with the stellar classification

- ☐ Cecilia Payne
- ☐ Edward Pickering



Meghnad Saha



Annie Jump Cannon



Question 3 10 pts

Who claimed that stars are mostly made of hydrogen in his/her PhD thesis



Max planck



Annie Jump Cannon



Cecilia Paybe



Meghnad Saha



Question 4 10 pts

go to:

https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_all.html 

[\(https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_all.html\)](https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_all.html)

Upper right . click on everything. By default, you have a Sun. Its peak color is the green. On the right side there is a "thermometer". Decrease the temperature until the peak of the curve is in the red (red star). Its temperature is about:



about 4000K (6740F)



about 10,000K (17540F)



about 2000K (3000F)



about 6,000K (10340 F)



Question 5 10 pts

Use the same App. As you increase the temperature, the area under the graph (in grey)



decreases



increases



stays the same



Question 6 10 pts

The Luminosity of a star is its



power output (energy/s)



relative size to our Sun



relative amount of hydrogen



peak wavelength



Question 7 10 pts

Annie classified the stars based on (see slides)



distance from earth



their absorption lines



hydrogen content



luminosity



Question 8 10 pts

Which is the right order for stellar classes;



O, B, A, F, G , K, M, T , L



O, B, A, F, G , K, Y, L, T



O, B, A, F, G , K, M, L, T



K, B, G, F,A , O, M, L, T



Question 9 10 pts

slide 20. Wien found a relationship between the peak wavelength of a blackbody (like a star, oven). As the peak wavelength increases, the temperature



decreases

☐

no relationship

☐

insreases



Question 10 10 pts

slide 25. Cecilia Payne improved the classification of stars. Each class now has a typical "QR code" (absorption lines) with a matching temperature and color. She based her finding on the work done by Saha and Wien.

..... stars exhibit absorption lines from transitions in neutral metal atoms and molecular bands, like titanium oxide, due to their cooler temperatures enabling molecule formation.

☐

F

☐

A

☐

M

☐

O

☐

G



Question 11 5 pts

bonus

Helium was discovered by observing the spectrum of

☐

Sirius A

☐

the Sun

☐

the Andromeda galaxy

☐

hot plasma lamp

Quiz saved at 4:38pm

Submit Quiz